

Final Project Report: How Music Taste Changes Over Time

Lauren Yee
Kian Sharifzadeh
Krish Doshi

Stats 423: Applied Linear Regression & ANOVA

March 10, 2026

Abstract

Music is a key part of daily life, yet people's opinions about music have evolved over time. With factors such as race having a different impact on society during each decade, this study focuses on how various factors affect song ratings during different decades. Data from Billboard Hot 100 Number One songs were taken from the 1960s – 2010s and split up to form rating prediction linear models. We compared main effects models with interaction effects models and decided which one would be more viable to optimize. The chosen model for the decade was optimized based on its AIC score to determine which predictors made a significant impact on the model. Comparing the predictors across models revealed that variables such as energy and race were consistently important across most decades. However, no singular factor was consistent between every decade. Ultimately, the findings demonstrated that while the factors that predict song ratings change over time, they have the potential to be similar in different time periods.

1 Introduction

Research Questions

1. What are the most and least optimal predictors over time from 1960–2010?
2. How do the models stay consistent and how do they change?
3. How does the predictive power of predictors change over time and which model performs best?

Our interest in music inspired our group to investigate its distinction over time. This prompted dividing our data into different decades to group similar song eras together, allowing us to understand what characteristics truly impact the overall rating. Since music is vastly different in each generation, it is interesting to see what aspects of music and the artist predict how a song is rated. This study shows the historical differences between predictors and how aspects such as trends and race can affect how a song is rated.

2 Data Description and Exploratory Data Analysis (EDA)

Data Description

This dataset includes information about every Number 1 song on the Billboard charts from 1958–2025. It was compiled by Chris Dalla Riva, who listened to each number one hit song and tracked data related to each song. Most of the data comes from facts about the song, Chris’s interpretation about the song, or data and ratings from Spotify.

2.1 Variables

Let Y denote the overall rating of the song.

- X_1 : Age: Average age of the artist(s)
- X_2 : GenderNotMale: Whether any of the artists aren’t male (1 if not male, 0 otherwise)
- X_3 : RaceWhite: Whether any of the artists are white (1 if white, 0 otherwise)
- X_4 : GenreJazz: Whether the song is jazz (1 if jazz, 0 otherwise)
- X_5 : GenreRock: Whether the song is rock (1 if rock, 0 otherwise)
- X_6 : GenrePop: Whether the song is pop (1 if pop, 0 otherwise)
- X_7 : GenreOther: Whether the song is a different genre (1 if not rock, pop, jazz or country and 0 otherwise)
- X_8 : Energy: Energy measure from Spotify (0-100)
- X_9 : Danceability: Danceability measure from Spotify (0-100)
- X_{10} : Happiness: Happiness measure from Spotify (0-100)
- X_{11} : Loudness_db: Loudness measured in decibels from Spotify

- X_{12} : BPM: Beats per minute from Spotify
- X_{13} : Length_sec: Length of the song in seconds

2.2 Exploratory Analysis

First we looked at the numeric predictors: energy, danceability, happiness, loudness_db, bpm, length_sec, and age.

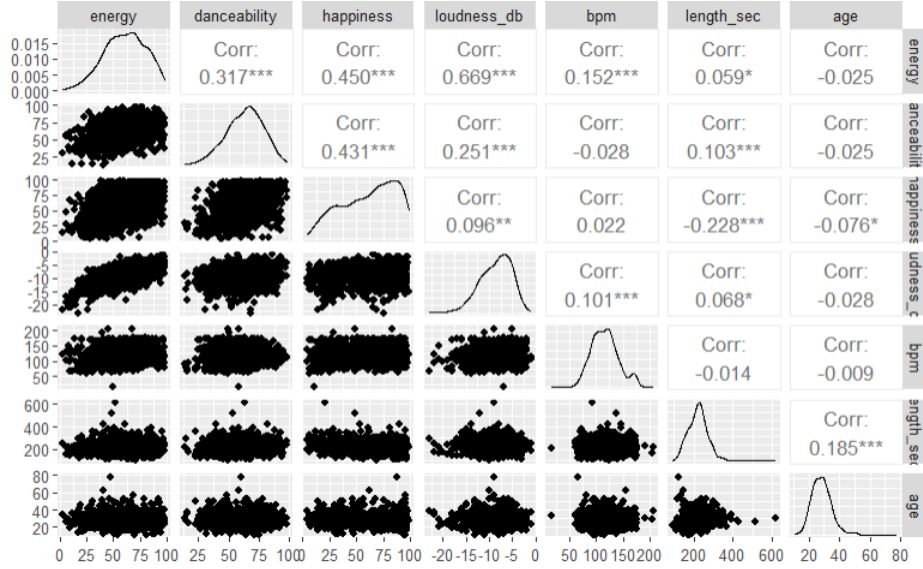


Figure 1: Scatterplot Matrix for Numeric Predictors

3 Methodology and Model Building

Model Fitting Procedure

1. Fit a main effects and interaction effects model to predict the decade's overall song ratings using the variables states above
2. Determine whether the interaction effects model included substantial predictors that weren't included in our main effects model by running an anova test and interpreting the p-value (at a significance level of 0.1)
3. If the p-value was significant, then run a backwards step procedure on the interaction effects model to remove insignificant predictors based on the change in AIC per step. Otherwise, run the backwards step procedure on the main effects model.
4. Repeat the previous steps for each decade to acquire the optimized models

1960s Model

$$\begin{aligned} Y_i = & \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \beta_4 X_{i4} + \beta_5 X_{i5} \\ & + \beta_6 X_{i6} + \beta_7 X_{i7} + \beta_8 X_{i8} + \beta_9 X_{i9} + \beta_{10} X_{i10} + \beta_{11} X_{i13} \\ & + \beta_{12}(X_{i3}X_{i4}) + \beta_{13}(X_{i3}X_{i7}) + \beta_{14}(X_{i3}X_{i6}) \\ & + \beta_{15}(X_{i3}X_{i8}) + \beta_{16}(X_{i3}X_{i9}) + \beta_{17}(X_{i2}X_{i9}) \\ & + \beta_{18}(X_{i2}X_{i10}) + \epsilon_i \end{aligned} \quad (1)$$

1970s Model

$$\begin{aligned} Y_i = & \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i3} + \beta_3 X_{i4} + \beta_4 X_{i5} \\ & + \beta_5 X_{i6} + \beta_6 X_{i7} + \beta_7 X_{i13} + \epsilon_i \end{aligned} \quad (2)$$

1980s Model

$$Y_i = \beta_0 + \beta_1 X_{i8} + \beta_2 X_{i9} + \beta_3 X_{i10} + \epsilon_i \quad (3)$$

1990s Model

$$Y_i = \beta_0 + \beta_1 X_{i2} + \beta_2 X_{i3} + \beta_3 X_{i8} + \beta_4 X_{i10} + \beta_5 X_{i11} + \epsilon_i \quad (4)$$

2000s Model

$$\begin{aligned} Y_i = & \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \beta_4 X_{i5} \\ & + \beta_5 X_{i6} + \beta_6 X_{i7} + \beta_7 X_{i8} + \beta_8 X_{i9} + \beta_9 X_{i10} + \beta_{10} X_{i12} \\ & + \beta_{11} X_{i13} + \beta_{12}(X_{i1}X_{i9}) + \beta_{13}(X_{i3}X_{i7}) \\ & + \beta_{14}(X_{i3}X_{i9}) + \beta_{15}(X_{i3}X_{i10}) + \beta_{16}(X_{i3}X_{i12}) \\ & + \beta_{17}(X_{i3}X_{i13}) + \beta_{18}(X_{i2}X_{i9}) + \beta_{19}(X_{i2}X_{i10}) \\ & + \beta_{20}(X_{i2}X_{i12}) + \epsilon_i \end{aligned} \quad (5)$$

2010s Model

$$Y_i = \beta_0 + \beta_1 X_{i2} + \beta_2 X_{i8} + \beta_3 X_{i10} + \beta_4 X_{i13} + \epsilon_i \quad (6)$$

3.1 Proposed Model

We chose the model that best explained the variability in overall ratings in its decade, while still penalizing complexity (best adjusted R^2 value). According to the adjusted r^2 values, the linear model predicting overall song ratings in the 60s was the best model.

Therefore, we fit a multiple linear regression model of the form:

$$\begin{aligned}
Y_i = & \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \beta_4 X_{i4} + \beta_5 X_{i5} \\
& + \beta_6 X_{i6} + \beta_7 X_{i7} + \beta_8 X_{i8} + \beta_9 X_{i9} + \beta_{10} X_{i10} + \beta_{11} X_{i13} \\
& + \beta_{12}(X_{i3}X_{i4}) + \beta_{13}(X_{i3}X_{i7}) + \beta_{14}(X_{i3}X_{i6}) \\
& + \beta_{15}(X_{i3}X_{i8}) + \beta_{16}(X_{i3}X_{i9}) + \beta_{17}(X_{i2}X_{i9}) \\
& + \beta_{18}(X_{i2}X_{i10}) + \epsilon_i
\end{aligned} \tag{7}$$

where $\epsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$.

Explain how you selected your variables (e.g., stepwise selection, theoretical justification, handling of multicollinearity). If you used transformations (like taking the log of the response), explain why.

4 Results and Diagnostics

Decade	Adjusted R^2
1960s	0.245
1970s	0.201
1980s	0.010
1990s	0.215
2000s	0.220
2010s	0.064

Table 1: Adjusted R^2 for Each Optimized Model Per Decade

4.1 Model Summary

Since the 1960s model had the highest adjusted R^2 value, we will use it as our final model

```

=====
SUMMARY FOR OPTIMAL MODEL: 1960s
=====

Call:
lm(formula = rating ~ age + gender + race + genre + energy +
    danceability + happiness + length_sec + race:genre + race:energy +
    race:danceability + gender:danceability + gender:happiness,
    data = subpop)

Residuals:
    Min       1Q   Median       3Q      Max
-5.3064 -1.2960  0.1313  1.2436  4.2045

Coefficients: (1 not defined because of singularities)
            1          Estimate Std. Error t value Pr(>|t|)
(Intercept) 6.235892    2.001360   3.116  0.00214 **
age         -0.036813    0.021122  -1.743  0.08309 .
genderNot Male -1.820687    1.293523  -1.408  0.16101
raceWhite    1.604179    1.795663   0.893  0.37287
genreJazz    1.473039    1.332354   1.106  0.27040
genreOther   1.978211    0.999147   1.980  0.04926 *
genrePop     0.988531    0.889417   1.111  0.26788
genreRock    0.260014    0.629236   0.413  0.67994
energy       -0.005765    0.015336  -0.376  0.70743
danceability -0.013030    0.022465  -0.580  0.56264
happiness     0.004845    0.009013   0.538  0.59154
length_sec   0.008965    0.003791   2.365  0.01912 *
raceWhite:genreJazz -1.353373    2.222521  -0.609  0.54334
raceWhite:genreOther -3.249204    1.187123  -2.737  0.00683 **
raceWhite:genrePop -1.786329    0.784348  -2.277  0.02395 *
raceWhite:genreRock      NA           NA      NA      NA
raceWhite:energy    0.024399    0.017347   1.407  0.16132
raceWhite:danceability -0.046982    0.022464  -2.091  0.03791 *
genderNot Male:danceability 0.076584    0.023552   3.252  0.00137 **
genderNot Male:happiness -0.030694    0.016106  -1.906  0.05830 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.797 on 178 degrees of freedom
Multiple R-squared:  0.314,    Adjusted R-squared:  0.2446
F-statistic: 4.527 on 18 and 178 DF,  p-value: 4.551e-08

```

Figure 2: Summary of 1960s Optimized Model

4.2 Model Diagnostics

The figures below represent the TA, QQ-plots, and Residuals vs Leverage for our selected optimized model from the 1960s

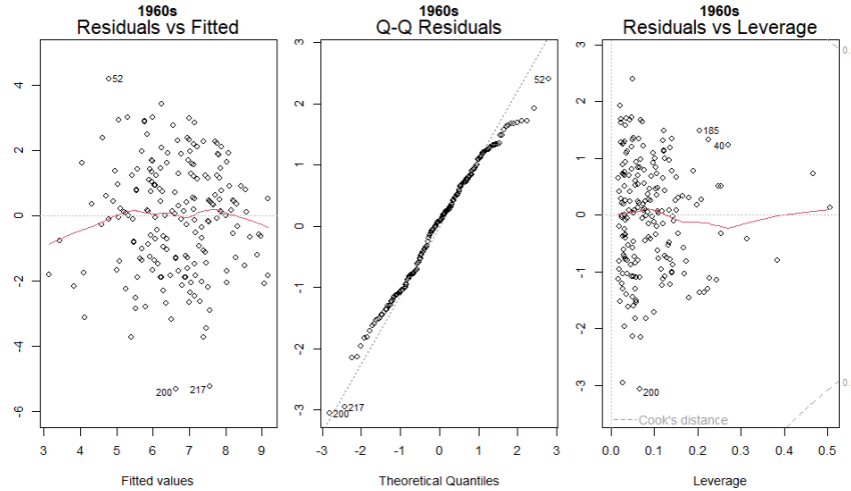


Figure 3: TA and QQ Plot for the 1960s Model

In the TA plot, we see that the residuals are mostly around 0, which justifies the zero mean assumption. Our observations on the TA plot illustrate that the residuals, justifying the homoscedasticity assumption. With the QQ plot, we noticed that the tails are somewhat

short-tailed on both ends for our final model. Because of this, we violate the normality assumption. However, since the short tails are without skew in the QQ plot, we believed that transformations were unnecessary for the final model. The residuals vs. leverage show that all of our observations have a Cook's distance of under 0.5, meaning that there are not significant enough outliers to perform transformations.

5 Discussion and Conclusion

Summarize the main takeaways from your model in the context of the original problem. Discuss coefficient interpretation and study limitations.

References

References

- [1] Chris Dalla Riva. *Billboard Hot 100 Number Ones*. <https://github.com/rfordatascience/tidytuesday/tree/main/data/2025/2025-08-26>

A Appendix: Code

```
# Example R Code
model <- lm(Y ~ X1 + X2, data = mydata)
summary(model)
plot(model)
```

A.1 Cleaning Data

```
orig_data <- read.csv("billboard.csv")

cleaned_data <- data.frame(artist = orig_data$artist, rating = orig_data$overall_rating,
  age=orig_data$front_person_age)

cleaned_data$gender <- ifelse(orig_data$artist_male == 1, "Male", "Not Male")

cleaned_data$race <- ifelse(orig_data$artist_white == 1, "White", "Not White")

cleaned_data$genre <- ifelse(is.na(orig_data$cdr_genre), "Genre Not Specified",
  orig_data$cdr_genre)

cleaned_data$genre <- ifelse(grepl("Pop", orig_data$cdr_genre), "Pop",
  ifelse(grepl("Country", orig_data$cdr_genre), "Country",
  ifelse(grepl("Hip-Hop", orig_data$cdr_genre), "Hip-Hop",
  ifelse(grepl("Rock", orig_data$cdr_genre), "Rock",
  ifelse(grepl("Jazz", orig_data$cdr_genre), "Jazz", "Other"))))))

cleaned_data <- cbind(cleaned_data, bpm=orig_data$bpm, energy=orig_data$energy,
  danceability = orig_data$danceability, happiness = orig_data$happiness,
  loudness_db = orig_data$loudness_d_b,
  length_sec = orig_data$length_sec)

cleaned_data$year <- as.numeric(substr(orig_data$date, 1,
  unlist(gregexpr("-", orig_data$date))[1] - 1))

cleaned_data <- cleaned_data[rowSums(is.na(cleaned_data)) == 0 , ]
```


A.2 Finding Optimized Models

```
model_results <- list()

for(i in seq(1960, 2030, 10)){
  decade_label <- paste0(i - 10, "s")

  subpop <- cleaned_data[cleaned_data$year >= i - 10 &
                        cleaned_data$year < i, ]

  # --- SAFETY CHECK 1: Ensure enough rows exist ---
  # If you have < 50 rows, the interaction model will likely fail.
  if(nrow(subpop) < 50) {
    message(paste("Skipping", decade_label, "- Not enough data points."))
    next
  }

  factor_counts <- sapply(subpop[, c("genre", "gender", "race")],
                        function(x) length(unique(na.omit(x))))

  if(any(factor_counts < 2)) {
    message(paste("Skipping", decade_label, "due to single-level factor:",
                  names(factor_counts)[factor_counts < 2]))
    next
  }

  first_model <- lm(data = subpop, rating ~ age + race + gender + genre + energy
                    + danceability + happiness + bpm + loudness_db + length_sec)

  second_model <- lm(data = subpop, rating ~ age + gender + race + age*genre
                    + age*energy + age*danceability + age*happiness + age*bpm
                    + age*loudness_db + age*length_sec + race*genre + race*energy
                    + race*danceability + race*happiness + race*bpm + race*loudness_db
                    + race*length_sec + gender*genre + gender*energy + gender*danceability
                    + gender*happiness + gender*bpm + gender*loudness_db + gender*length_sec)
```

```

if(anova(first_model, second_model)$"Pr(>F)"[2] <= 0.1){
  optimal_model <- step(second_model, direction="backward")
} else {
  optimal_model <- step(first_model, direction="backward")
}

model_results[[decade_label]] <- list(
  "first"    = first_model,
  "second"   = second_model,
  "optimal"  = optimal_model
)

print(decade_label)
summary(first_model)
summary(second_model)
summary(optimal_model)
}

```

A.3 Response Analysis

```

ggplot(cleaned_data, aes(x = rating)) +
  geom_histogram(
    binwidth = 1,
    boundary = 0.5,
    fill = "black",
    color = "grey",
    linewidth = 0.3
  ) +
  labs(
    title = "Histogram of Song Ratings",
    x = "Rating",
    y = "Frequency"
  ) +
  theme_classic()

```

A.4 Adjusted R^2 Comparison

```
summary(model_60s)$adj.r.squared

summary(model_70s)$adj.r.squared

summary(model_80s)$adj.r.squared

summary(model_90s)$adj.r.squared

summary(model_2000s)$adj.r.squared

summary(model_2010s)$adj.r.squared

library(ggplot2)

df <- data.frame(
  Decade = factor(c("1960s", "1970s", "1980s", "1990s", "2000s", "2010s"),
    levels = c("2010s", "2000s", "1990s", "1980s", "1970s", "1960s")),
  Adj_R2 = c(0.244647, 0.2013403, 0.009698796, 0.2152039, 0.2197517, 0.06362834)
)

ggplot(df, aes(x = Decade, y = Adj_R2)) +
  geom_col(fill = "#23DC6D") +
  geom_text(aes(label = round(Adj_R2, 3)), hjust = -0.1, size = 4) +
  coord_flip() +
  ylim(0, 0.3) +
  labs(
    title = "Adjusted R2 of Song Rating Models by Decade",
    x = "Decade",
    y = "Adjusted R2"
  ) +
  theme_minimal(base_size = 14)
```